

Use Case Description

Workout-Management-System

ID and name	UC-1: Authenticate User		
Primary actor	User	Secondary actors	Authentication System
Description	User authenticates in the workout management application.		
Trigger	User intends to authenticate in the application.		
Preconditions	PRE-1. User is registered in the system.		
Postconditions	POST-1. The system displays a personalized version of the application for the authenticated user.		
Normal flow	1.0 Authenticate 1. User enters the login credentials and tries to authenticate. 2. Authentication System recognizes the provided data and opens a new personalized dashboard for User. The authentication succeeded. (see 1.0.E1)		
Alternative flows	None		
Exceptions	1.0.E1 The authentication details entered by User are not valid (e.g. the password is incorrect or the account does not exist). 1. Authentication System informs User about the situation with a suggestive message. The use case restarts.		

ID and name	UC-2: Register User		
Primary actor	User	Secondary actors	Authentication System
Description	User creates a new user account for the workout management application.		
Trigger	User intends to create a new account.		
Preconditions	PRE-1. User does not have another account linked to new account data.		
Postconditions	POST-1. A new user account is stored in the Authentication System.		

Normal flow	1.0 Create a new user account 1. User asks to create a new account. 2. Authentication System provides a registering form to complete. 3. User fills in at least all the mandatory sections. 4. Authentication System gathers all the information provided and stores a new user account. (see 1.0.E1) 5. Authentication System confirms the action and redirects User back to authentication section.
Alternative flows	None
Exceptions	1.0.E1 Data provided by User is not valid. For instance, the email address is already linked to another account, or not all the mandatory fields were filled in. 1. Authentication System informs User about the situation with a suggestive message. The use case restarts.

ID and name	UC-3: Manage workout plans		
Primary actor	User	Secondary actors	Workout Management System (WMS)
Description	User wants to view all existing workout plans linked to him or add an empty workout plan or modify/delete an existing one.		
Trigger	User tries to access his workout plans.		
Preconditions	PRE-1. User is registered in WMS.		
Postconditions	POST-1. If it was created: WMS stores a new empty workout plan. POST-2. If it was updated: WMS modifies the information of the selected workout plan. POST-3: If it was deleted: WMS removes permanently the workout plan.		
Normal flow	1.0 View workout plans 1. User tries to access his workout plans. 2. WMS displays all the workout plans linked to User. 2.0 Create new workout plan 1. User requests to add a new workout plan. 2. WMS provides User the data he needs to fill in. 3. User sends the information. 4. WMS responds by storing a new empty workout plan having those properties. (see 2.0.E1) 5. WMS updates the list of workout plans displayed. 3.0 Update workout plan		

	1. User selects a workout plan. 2. WMS displays all the information about it. 3. User can decide to change its attributes or add, modify or remove a day from it. (see 3.2) 4. WMS updates the workout plan stored and the information displayed. (see 3.0.E1) 5. User asks to leave the section about the workout plan modified. (see 3.1) 6. WMS acts accordingly, changing the view to what was before. 4.0 Delete workout plan 1. User selects a workout plan to remove. 2. WMS displays a confirmation message to User. 3. User confirms the deletion. (see 4.1) 4. WMS deletes the workout plan from the storage unit and updates the display.
Alternative flows	3.1 User decides to keep modifying the workout plan 1. Return to step 3 of normal flow. 3.2 User leaves without making any changes 1. Jump to step 6 instantly. 4.1 User has changed his mind and decides not to delete the workout plan 1. User declines the confirmation message. 2. WMS terminates the use case.
Exceptions	2.0.E1 The specifications provided related to the workout plan are not complete. 1. WMS informs User about the problem that had occurred. 2a. User decides to start the process again. The use case restarts. 2b. User chooses to cancel the adding process. WMS terminates the use case. 3.0.E1 The new properties provided by User are not complete. 1. WMS informs User about the problem that had occurred. The use case goes back to step 3. However, all the changes made to the workout days persists.

ID and name	UC-4: Manage workout plan days		
Primary actor	User	Secondary actors	Workout Management System (WMS)

Description	User wants to view all existing workout days of the selected workout plan or add an empty day to the plan or modify/delete an existing one.
Trigger	User selects a workout plan.
Preconditions	PRE-1. User is registered in WMS. PRE-2. User has selected to view a workout plan.
Postconditions	POST-1. If it was created: WMS stores a new empty workout day. POST-2. If it was updated: WMS modifies the information of the selected workout day. POST-3: If it was deleted: WMS removes permanently the workout day from the plan.
Normal flow	1.0 View workout day 1. User selects a workout day from the plan and requests to see the information stored. 2. WMS displays all its details. 2.0 Create a new workout day 1. User asks to add a new empty workout day for the current workout plan selected. 2. WMS displays a form for User to fill in. 3. User provides the necessary data. 4. WMS gathers the information and stores the workout day linked to the selected workout plan. (see 2.0.E1) 5. WMS updates the list of displayed workout days. 3.0 Update workout day 1. User selects a workout day. 2. WMS displays all its details. 3. User can decide to modify its attributes or add a new exercise or modify the information about an existing one or delete it. (see 3.2) 4. WMS updates the details of the workout day and updates the displayed information about the workout plan. (see 3.0.E1) 5. User asks to end the section related to updating the workout day. (see 3.1) 6. WMS terminates the use case. 4.0 Delete a workout day 1. User selects a workout day and requests to delete it. 2. WMS sends a confirmation message to User. 3. User confirms its decision. (see 4.1) 4. WMS deletes the workout plan forever and updates the displayed information about the workout plan.
Alternative flows	3.1 User decides to keep modifying the workout plan 1. Return to step 3 of normal flow.

	3.2 User leaves without making any changes 1. Jump to step 6 instantly. 4.1 User has changed his mind and decides not to delete the workout plan 1. User declines the confirmation message. 2. WMS terminates the use case.
Exceptions	2.0.E1 The specifications provided related to the workout day are not complete. 1. WMS informs User about the problem that had occurred. 2a. User decides to start the process again. The use case restarts. 2b. User chooses to cancel the adding process. WMS terminates the use case. 3.0.E1 The new properties provided by User are not complete. 1. WMS informs User about the problem that had occurred. The use case goes back to step 3. However, all the operations made on the exercises persists.

ID and name	UC-5: Manage exercises from workout day		
Primary actor	User	Secondary actors	Workout Management System (WMS)
Description	For a selected workout day from a workout plan, the user can choose to either add a new exercise, modify an existing exercise's content or delete it.		
Trigger	User selects a workout day and an exercise from it or decides to add a new exercise.		
Preconditions	PRE-1. User is registered in WMS. PRE-2. User has selected to view a workout plan. PRE-3. User has selected a workout day from the workout plan.		
Postconditions	POST-1. If it has been added: WMS stores a new workout exercise linked to the workout day of the workout plan. POST-2. If it has been updated: WMS modifies the information about the selected exercise.		

	POST-3. If it has been deleted: WMS removes permanently the exercise from the workout day.
Normal flow	1.0 User adds or modifies an exercise in the workout plan 1. User asks to add or modify an exercise on the workout day. 2. WMS displays a form containing relevant information for an exercise, such as weight used, number of sets/reps, pause between sets, duration etc. 3. User fills in the form. 4. WMS collects the data and stores/modifies an exercise on the workout day. (see 1.0.E1) 5. WMS updates the displayed details about the workout day selected. 2.0 View an exercise from a workout day 1. User selects an exercise and asks to see its details. 2. WMS displays the requested information. 3.0 Delete an exercise from a workout day 1. User selects an exercise and requests to remove it from the workout day. 2. WMS deletes the exercise from the workout plan forever, but it can easily be added back later.
Alternative flows	None
Exceptions	1.0.E1 The specifications provided related to the exercise are not complete. 1. WMS informs User about the problem that had occurred. 2a. User decides to start the process again. The use case restarts. 2b. User chooses to cancel the adding/updating process. WMS terminates the use case.

ID and name	UC-6: Manage exercises		
Primary actor	User	Secondary actors	Workout Management System (WMS)
Description	User wants to see all the existing exercises. Initially, all of them are predefined. However, User can create custom exercises (later he can update or delete them). Custom exercises are visible only to the user that created them. Selecting an exercise leads to seeing its details.		
Trigger	User wants to access all the existing exercises and selects one of them or decides to add a new one.		
Preconditions	PRE-1. User is registered in WMS.		
Postconditions	POST-1. If it has been created: WMS stores a new custom exercise.		

	POST-2. If it has been updated: WMS modifies the information of the selected custom exercise. POST-3. If it has been deleted: WMS removes permanently the custom exercise selected.
Normal flow	1.0 View exercises 1. WMS displays all the exercises (system exercises or made by User). 2. User selects an exercise from the collection. 3. WMS displays its details. 2.0 Create custom exercise 1. User requests to add a new custom exercise. 2. WMS displays a form containing all the relevant data related to the exercise: the main muscle group involved, secondary muscle group involved (up to 2), exercise type, name, equipment used (up to 2). 3. User fills in the necessary data. 4. WMS gathers the information and stores the new custom exercise and links it to User. (see 2.0.E1) 5. WMS updates the exercises collection displayed to User. 3.0 Update custom exercise 1. User selects a custom exercise and requests to modify its details. 2. WMS displays an already completed form containing the same information as when adding an exercise 3. User modifies the values of the fields he desires. 4. WMS gathers the new exercise version and replaces the old one stored. (see 3.0.E1) 5. WMS updates the information displayed and goes back to step 3 until User decides to give up. 4.0 Delete custom exercise 1. User selects a custom exercise and asks to remove it. 2. WMS sends a confirmation message to User. 3. User confirms his decision to delete the exercise. 4. WMS permanently deletes the custom exercise selected and updates the exercises displayed to User. 5.0 Filter exercises 1. User selects the filters to use. 2. WMS displays all the exercises after applying the filter.
Alternative flows	None
Exceptions	2.0.E1 The specifications provided related to the custom exercise are not complete or correct. 1. WMS informs User about the problem that had occurred. 2. The use case restarts.

	3.0.E1 The new attributes provided by User are not valid. He might have introduced wrong types or left some of them empty. 1. WMS informs User about the problem that had occurred. The use case goes back to step 3.
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ID and name	UC-7: Manage profile		
Primary actor	User	Secondary actors	Authentication System (AS)
Description	User wants to view his personal and account details and maybe modify them. He can also decide to delete his account.		
Trigger	User tries to access details about his account.		
Preconditions	PRE-1. User is registered in WMS.		
Postconditions	POST-1. If details have been modified: AS updates the old data regarding User account. POST-2. If account has been deleted: AS removes permanently the user account and logs out.		
Normal flow	1.0 View account details 1. User requests to see his personal and account details. 2. AS displays all the information. 2.0 Modify account details 1. User changes the value of at least one of the details. 2. AS gathers the new details version and replaces the old value. (see 2.0.E1) 3. AS updates the personal and account details displayed. 3.0 Delete account 1. User asks to delete his account. 2. AS sends a confirmation message to User. 3. User confirms his initial choice. (see 3.1) 4. AS performs a delete action for the account and logs out User, returning back to the authenticate state.		
Alternative flows	3.1 User has changed his mind and decides not to delete his account 1. User declines the confirmation message. 2. AS terminates the use case.		
Exceptions	2.0.E1 The new information is not valid. For instance, the confirm password and the password do not match (when User tries to change it) or some mandatory fields are left empty. 1. AS informs User about the problem that has occurred.		

	2. AS terminates the use case.
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ID and name	UC-8: View statistics and progress		
Primary actor	User	Secondary actors	Workout Management System (WMS)
Description	User can view statistical information about number of workouts or weight lifted in the last week/2 weeks or other predefined time span. Also, for each day of each month since the account has been created, User can see if he has done a workout and show the weight lifted, total workout time and number of workouts.		
Trigger	User requests to view statistical and progress related data about him.		
Preconditions	PRE-1. User is registered in WMS.		
Postconditions	None		
Normal flow	1.0 View total values for given period 1. User chooses a period to see the total weight lifted or number of workouts, total time spent. 2. WMS displays the requested value. 2.0 View progress for a month 1. User chooses a month since his account has been created until now to view for each day details like weight lifted or number of workouts, total time spent. 2. WMS displays the requested data.		
Alternative flows	None		
Exceptions	None		

ID and name	UC-9: Start workout session		
Primary actor	User	Secondary actors	Workout Management System (WMS)
Description	User can start a workout session. During the whole process, it shows the time passed since it began. By default, the order of the exercises is the one displayed, but User can choose to do any of the existent exercises of the workout day selected. After finishing a set for an exercise, a pause timer starts. User can decide to end the session at any point. At that moment WMS displays the total weight lifted and whole session duration.		

Trigger	User informs WMS that he intends to begin a workout session for a chosen workout day from a plan.
Preconditions	PRE-1. User is registered in WMS. PRE-2. User has selected a workout plan. PRE-3. User has selected a workout day from the workout plan.
Postconditions	POST-1: WMS stores a new workout session, initially marked as “in progress”. POST-2: At the end of the workout session, WMS updates it’s information and marks it as “done”.
Normal flow	1.0 Start a workout session 1. User requests to begin a new workout session for a workout day. 2. WMS sends a confirmation message to User to assure it was intentional. 3. User confirms his intention to start the session. (see 1.1) 4. WMS stores a new workout session, initially marked as “in progress”. 5. WMS displays information about the current session, such as elapsed time and the list of exercises. (see 1.4) 6. User performs a set for an exercise and marks it as done. (see 1.3, 1.5) 7. WMS starts a pause timer corresponding to the timer between sets of that exercise chosen by User. When the timer ends, jump back to step 5. (see 1.2)
Alternative flows	1.1 User has changed his mind and decides not to start a new workout session for the selected workout day. 1. User does not confirm his initial choice. 2. WMS terminates the use case. 1.2 User stops the timer before it ends. 1. User decides to stop the timer between sets prematurely. 2. WMS jumps back to step 5. 1.3 No more exercises can be performed by User. 1. User has exhausted all the sets and exercises of the workout session. 2. User requests to end the workout session. 3. WMS sends a confirmation message to User.

	4. User confirms his choice. 5. WMS ends the session and updates its value and marks it as “done”. 6. WMS displays an overall review of the workout session, including the total weight lifted and the time spent. 1.4 User decides to end the workout session before finishing all the exercises 1. User requests to end the workout session prematurely. 2. WMS jumps to step 3 of alternative flow 1.3. 1.5 Some exercises have a duration for a set 1a. If the timer has ended, User can mark the exercise as done and the use case jumps back to step 7. 1b.1 The timer has not ended, and User decides to mark the exercise as done. 1b.2 WMS jumps to step 7 of the use case.
Exceptions	None

ID and name	UC-10: Logout User		
Primary actor	User	Secondary actors	Authentication System (AS)
Description	User signs out of the application, terminating their active session.		
Trigger	The user interacts with a visual element to requests the action of logging out.		
Preconditions	PRE-1. User is registered in WMS.		
Postconditions	POST-1. User’s active session is terminated. POST-2. AS displays the initial login/welcome screen.		
Normal flow	1.0 Log out from the account 1. User requests to log out from the account. 2. AS displays a confirmation prompt. 3. User confirms the action. (see 1.1) 4. AS terminates the session and returns a success response.		

	5. AS clears the local session data from the device and redirects the User to the login screen.
Alternative flows	1.1 User has changed their mind and chooses not to log out. 1. User declines the confirmation message. 2. AS terminates the use case.
Exceptions	None

Descriptions of fields:

- **ID and name:** Title should be descriptive and should usually begin with a verb, e.g. order, calculate, input, etc. ID can have any format but must be unique among all use cases.
- **Primary actor:** Person that wishes to accomplish a goal through the use of the system. Only a single primary actor per use case.
- **Secondary actors:** Actors that have an interest in the completion of the goal but that do not directly interact with the system.
- **Description:** Concise description of the purpose of the use case.
- **Trigger:** Condition internal or external to the system that prompts the use case to start.
- **Preconditions:** Conditions that must be true before the use case starts. Each should be labeled with an ID unique to the use case.
- **Postconditions:** Conditions that must be true after the use case ends normally. Each should be labeled with an ID unique to the use case.
- **Normal flow:** Detailed step-by-step description of the logical flow of the use case. It should describe an explicit two way interaction, with the system prompting for input and the actor responding accordingly. Each step should be numbered.
- **Alternative flows:** Flows that achieve the same goal as the normal flow but are expected to be less common or lower priority.
- **Exceptions:** Conditions that result in the normal flow ending prematurely due to an unrecoverable condition in the system. The condition that causes the flow should be clearly stated, as should be any other decisions that the actor must make in this situation.